

Daily Logic Drill #1

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Truth Values & Statements · If-Then Structure · Converse vs Original · Basic Direct Proof.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Hypothesis, conclusion, converse. Pure recognition, no working.

1. Identify the hypothesis and the conclusion of: “If n is divisible by 10, then n is divisible by 5.”
2. State the converse of: “If $x = 3$, then $x^2 = 9$.”
3. True or false: “If a shape is a square, then it is a rectangle.”
4. True or false: “If a shape is a rectangle, then it is a square.”
5. State the converse of: “If n is even, then n^2 is even.”
6. True or false: “ $n^2 > 0$ for every integer n .”
7. In “If $x > 3$ then $x^2 > 9$ ”, which part is the hypothesis and which is the conclusion?
8. True or false: “If $a = b$, then $a^2 = b^2$.”
9. State the converse of: “If n is a multiple of 6, then n is a multiple of 2.”
10. True or false: “The converse of a true statement is always true.”

Section B Manipulation Drills

(≤50 s each)

Build the converse of a compound statement, or write a short direct proof.

1. Write the converse of: “If n is divisible by 4 and by 3, then n is divisible by 12.” State whether the original statement and its converse are both true.
2. Prove directly: “If n is an even integer, then n^2 is even.”
3. State the converse of: “If $x > 0$ and $y > 0$, then $xy > 0$.” Is the converse true? Justify your answer.

4. True or false, with a one-line reason: “If a number’s digits sum to a multiple of 3, then the number is a multiple of 3.”
5. Prove directly: “If n is an odd integer, then n^2 is odd.”
6. Consider: “If $x^2 = 9$ then $x = 3$.” Is this true? If false, give the value of x that breaks it.
7. State the converse of Pythagoras’ theorem (“if a triangle has a right angle, then $a^2 + b^2 = c^2$ for its side lengths, with c the longest”). Is this converse also a true geometric fact?
8. Prove directly: “If a and b are both even integers, then $a + b$ is even.”
9. True or false, with a one-line reason: “If p is a prime number and $p \neq 2$, then p is odd.”
10. State the converse of: “If $x = y$, then $x^3 = y^3$.” Is the converse true?

Section C Substitution & Structure

(≤90 s each)

Converses that hide a false step, proofs that need one extra insight, and your first multiple-choice question.

1. Prove directly that the square of any multiple of 3 is itself a multiple of 9.
2. Prove directly: “If n is an odd integer, then $n^2 - 1$ is divisible by 8.”
3. Which of the following is the correct converse of: “If x is rational and y is irrational, then $x + y$ is irrational”?
 - A) If $x + y$ is irrational, then x is rational and y is irrational.
 - B) If x is irrational and y is rational, then $x + y$ is irrational.
 - C) If $x + y$ is rational, then x is irrational or y is rational.
 - D) If x is rational and y is rational, then $x + y$ is irrational.
4. True or false, with justification: “If a statement is true, its converse must also be false.”
5. Prove directly: “If n is divisible by 6, then n^3 is divisible by 216.”
6. State the converse of: “For positive reals x, y : if $x \geq 2$ and $y \geq 2$, then $xy \geq x + y$.” Determine whether the converse is true, giving a counterexample if it is false.
7. State the converse of: “If x and y are both perfect squares, then xy is a perfect square.” Is the converse true? Give a counterexample if it is false.
8. True or false, with justification: “If n^2 is a multiple of 4, then n is a multiple of 4.”

Section D Challenge

(TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. Let n be a positive integer. Consider the statements:

- I. If n is divisible by 4, then n is divisible by 8.
- II. If n is divisible by 8, then n is divisible by 4.
- III. The converse of statement I is statement II.

Which of the statements I, II, III are true?

- A) I only
- B) II only
- C) III only
- D) I and II only
- E) I and III only
- F) II and III only
- G) I, II and III
- H) None of them

2. Prove that if n and $n + 2$ are both prime numbers greater than 3, then $n + 1$ is divisible by 6.

3. Amy claims: “If a positive integer is divisible by every one of 2, 3, 4, 5, 6, then it is divisible by 60.” Determine whether Amy’s claim is true, and justify your answer using the smallest positive integer for which it can be checked.

4. Give an example of a true statement S about divisibility of integers whose converse S' is also true. State both S and S' .

5. Determine, with proof, whether the following statement is true or false: “If $n^2 - n$ is even, then n is even.”

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths